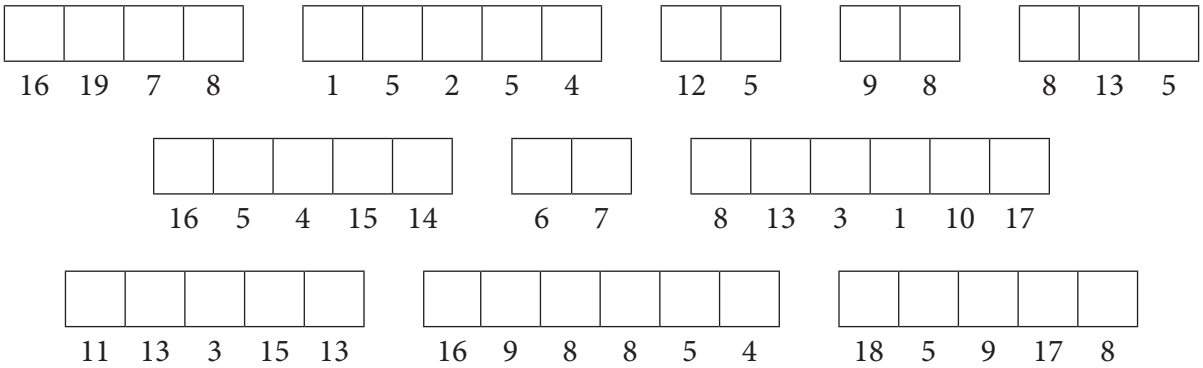
WORKSHEET

Chapter 7: Differentiating exponential and logarithmic functions

Things which matter most …



Goethe – poet and author

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Questions

Differentiate the following exponential and logarithmic functions.

1 $\ln(5x)$

2 $5\ln(8x)$

3 $7\ln(x)$

4 $\ln(x^3)$

5 $\ln(3x^4)$

6 $\ln[\ln(x)]$

7 $8\ln(4x)$

8 $\ln[\sin(x)]$

9 $\frac{1}{2}\ln(7x)$

10 $\ln(3x^2 + 4x - 2)$

11 $\ln(x^2 + 3x)$

12 $\ln(x^5 - 3x^2)$

13 $\ln[\cos(x)]$

14 $e^{\ln(x)}$

15 $\ln(xe^x)$

16 $2\ln(x^4 - \frac{3}{2}x)$

17 $5e^{3x} + \ln x$

18 $6e^{(\frac{1}{3}x^2 - 4)} - 2\ln(x^2)$

19 $\ln(x^3 + 5x) + 4e^x$

Solutions

A $\frac{1}{2x}$

B $\frac{5x^4 - 6x}{x^5 - 3x^2}$

C 1

E $\frac{4}{x}$

F $\frac{8}{x}$

G $\frac{6x + 4}{3x^2 + 4x - 2}$

H $-\tan(x)$

I $\frac{7}{x}$

L $4xe^{(\frac{1}{3}x^2 - 4)} - \frac{2}{x}$

M $\frac{8x^3 - 3}{x^4 - \frac{3}{2}x}$

N $\frac{1}{x}$

O $\frac{1}{x\ln(x)}$

R $\frac{3}{x}$

T $\cot(x)$

U $\frac{3x^2 + 5}{x^3 + 5x} + 4e^x$

V $\frac{5}{x}$

W $\frac{2x + 3}{x^2 + 3x}$

Y $\frac{1}{x} + 1$

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